Robotics Tech

How to add Frames?

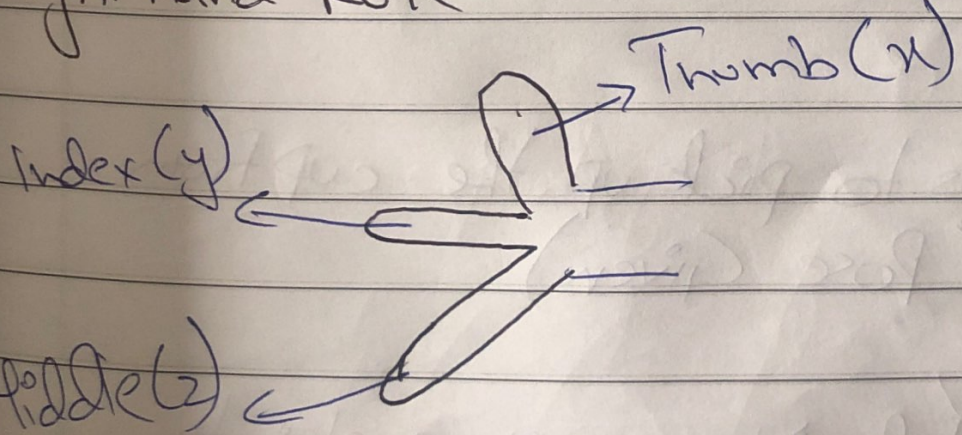
z-axis ⇒ Axis of rotation for revolute joint

x-axis ⇒ Perp to both current & previous z-axis

y-axis – Determined from x-axis & z-axis by using right hand rule

The x-axis must intersect the prev z-axis

Right hand Rule



↳ For Prismatic in the direction where it is moving.

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Inverse Kinematics

Given pose

Find the joint angles (Angle of each joint) to acheive that pose

$\xi_N = K(q)$ → joint angles given

- ξ_N ← Find Pose N
- $K(q)$ → Forward Kinematics

$q = K^{-1}(\xi_N)$ → Pose of N given

- q ← Find Angle q
- $K^{-1}(\xi_N)$ → Inverse Kinematics

Robot tries to pick up the cup from a table (Pose Given)

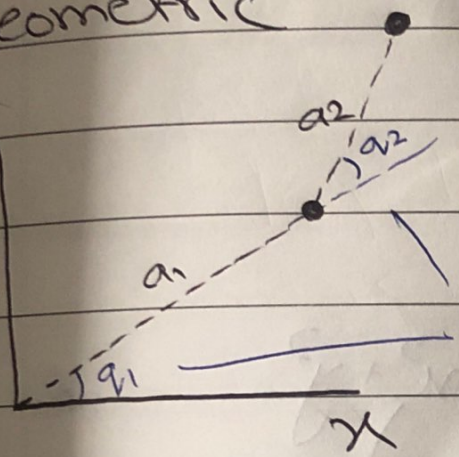
⇓

Calculate Necessary angles to reach & pick up the cup

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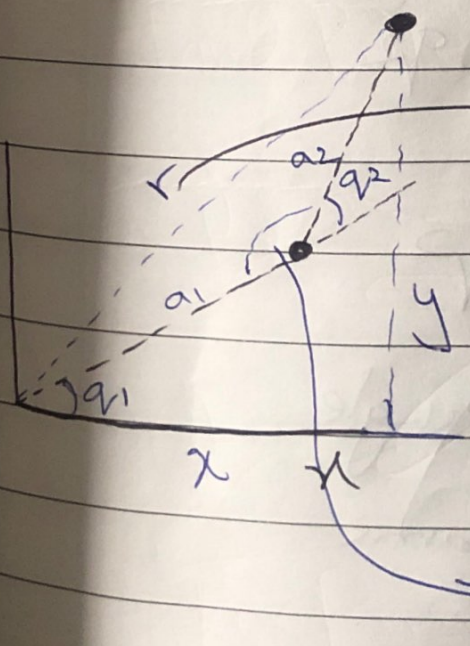
Inverse Kinematics Approach

1) Geometric



Use Geometric Approach to find

Make a Triangle Goal find q_1 & q_2



r find using cosine rule

$r^2 = a_1^2 + a_2^2 - 2(a_1)(a_2)\cos$

$\underline{(180 - q_2)}$

α

$\Rightarrow 180^\circ - q_2$

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We don't know value of α

Use formula

$$\cos\alpha = \frac{a_1^2 + a_2^2 - r^2}{2(a_1)(a_2)}$$

r^2 can be replaced by $x^2 + y^2$
(Pythagoras Theorem)
see Diagram
x, y is given

This will give us the value of α
Use that value to find q_2

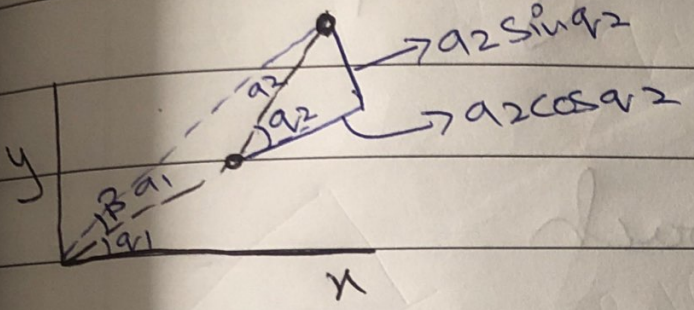
$q_2 = 180 - \alpha$

↳ We will get q_2

For q_1 use alternative angle Property

~~a~~ ~~a~~ ⇒ Same

Now lets find q_1



Lets find β first

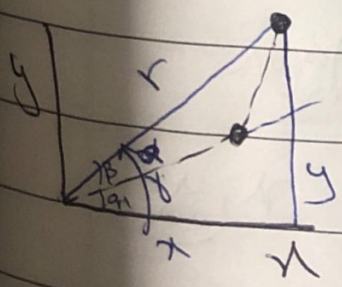
$$\tan\beta = \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

$$\beta = \tan^{-1}\left(\frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}\right)$$

⇓

This will give β

Now Cosider another triangle



Find γ now

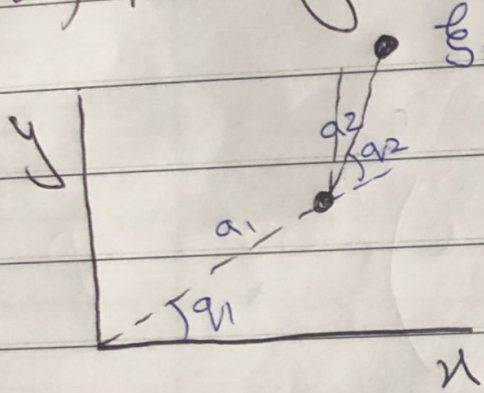
we know x & y

$$\gamma = \tan^{-1}\left(\frac{y}{x}\right)$$

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$q_1 = \gamma - \beta$ $\Rightarrow$ this will give q_1

2) Analytic Approach



$$E = R(q_1)\,T(a_1)\,R(q_2)\,T(q_2)$$

$$E = \begin{pmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & a_2\cos(q_1+q_2)+a_1\cos q_1 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & a_2\sin(q_1+q_2)+a_1\sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

Translation equal to x, y

Eq1

$x = a_2\cos(q_1+q_2) + a_1\cos q_1$

Eq2

$y = a_2\sin(q_1+q_2) + a_1\sin(q_1)$

To solve Simply Square & add

~~$x^2+y^2 = a_2^2\cos^2(q_1+q_2)+$~~

$x^2+y^2 = (a_2\cos(q_1+q_2)+a_1\cos q_1)^2$
$+ (a_2\sin(q_1+q_2)+a_1\sin(q_1))^2$

$= a_2^2\cos^2(q_1+q_2) + a_1^2\cos^2 q_1 + 2a_1a_2\cos(q_1+q_2)\cos q_1$
$+ a_2^2\sin^2(q_1+q_2) + a_1^2\sin^2 q_1 + 2a_1a_2\sin(q_1+q_2)\sin q_1$

$\cos^2\theta + \sin^2\theta = 1$

$\cos(q_1+q_2) = \cos q_1\cos q_2 - \sin q_1\sin q_2$

$\sin(q_1+q_2) = \sin q_1\cos q_2 + \cos q_1\sin q_2$

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This is just the derivation check from slides

We will use this formula to calculate joint angles

$$q_2 = \cos^{-1}\left(\frac{x^2+y^2-L_1^2-L_2^2}{2L_1L_2}\right)$$

$$q_1 = \tan^{-1}\left(\frac{y}{x}\right) - \tan^{-1}\left(\frac{L_2 \sin q_2}{L_1 + L_2 \cos q_2}\right)$$

L_1 & L_2 are the length of the link in this case its a_1 & a_2

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3) Numerical Approach ⇒ Based on guess

How far from the target position

→ Start with an initial guess

↳ Give random values to joint angles

→ Calculate current position

↳ Use forward kinematics to find the position of end effector after these angles

→ Determine the error

↳ Calculate how far are we from the goal

→ Adjust angles ⇒ Use optimization algorithm

→ Repeat until you reach your goal or get close enough

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Robot Motion, Path Trajectory

Joint-Interpolated Motion

↳ Move a robot smoothly from one pose (Position A) to another pose (Position B)

$$\xi_A \rightarrow \xi_B$$

Determine joint angles of each pose

↳ Interpolate b/w joint angles of initial & final positions

Example

Initial joint angles (Pos A)

$q_1 = [0, 0, 0, 0, 0]$ ↳ All joints at zero pos

Final joint Angles (Pos B)

$q_2 = [45, 30, 45, 0, 45]$

We have to go from $q_1 \rightarrow q_2$ smoothly

The path is not straight but curved

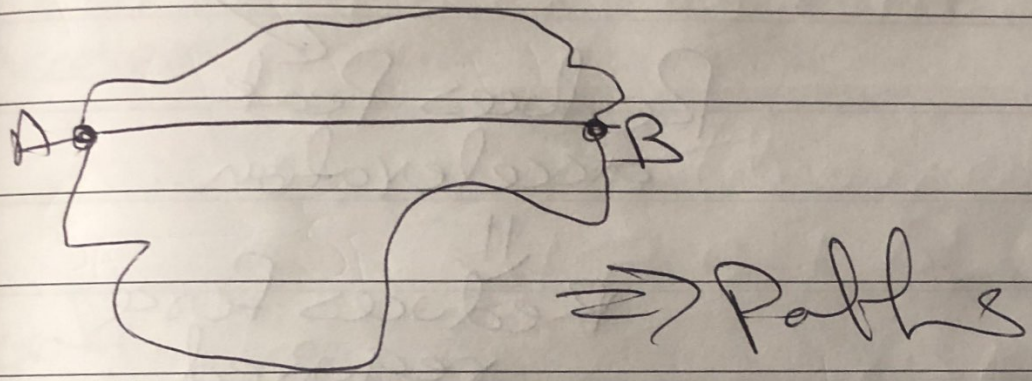
Path & Trajectories

Inertial Reference frame

- ~~Fixed~~ Does not accelerate or Rotate
  - ↓
  - Moves with constant velocity
- Interchangeably used with World Coordinate Frame
  - ↓
  - Frame fixed to some point on the Earth

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Path ——→ No time
- Geometric Description of Motion
- Move from one point to another



⇒ Paths

Trajectory
- Involves time
- Includes velocity and acceleration

Eg - Move from A to B in 2 seconds with a specific velocity

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Smoothness

└ Pose of robot vary smoothly over time

Reduces Peak acceleration

⇓

Reduces force required

⇓

Smaller Motors

⇓

Better Stability & Precision

Definition

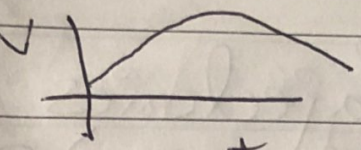
Position $x(t)$

Initial to final

↳ Must be continuous over time

No abrupt jumps or discontinuities in position

Velocity $\frac{dx}{dt}$

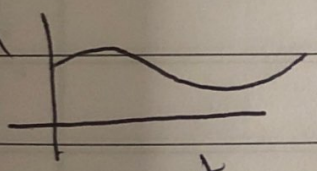


→ Velocity continuous

↓

Doesn't Accelerate or decelerate too quickly

Acceleration $\frac{d^2y}{dt^2}$



→ Continuous

↓

Smooth acceleration prevents jerky movements

Jerk $\frac{d^3y}{dt^2}$

→ Rate of change of acceleration ⇒ Continuous

↓

Should be minimized for → Smoothness

Smooth Trajectories

Quintic (5th Order) Polynomial

$$s(t) = At^5 + Bt^4 + Ct^3 + Dt^2 + Et + F$$

Boundary Conditions

Position at $t=0$ & $t=T$

Velocity " "

Acceleration " "

→ Equation for Position

$s'(t) \Rightarrow$ Eq for velocity

$s''(t) \Rightarrow$ Eq for acceleration

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Get values using Matrix

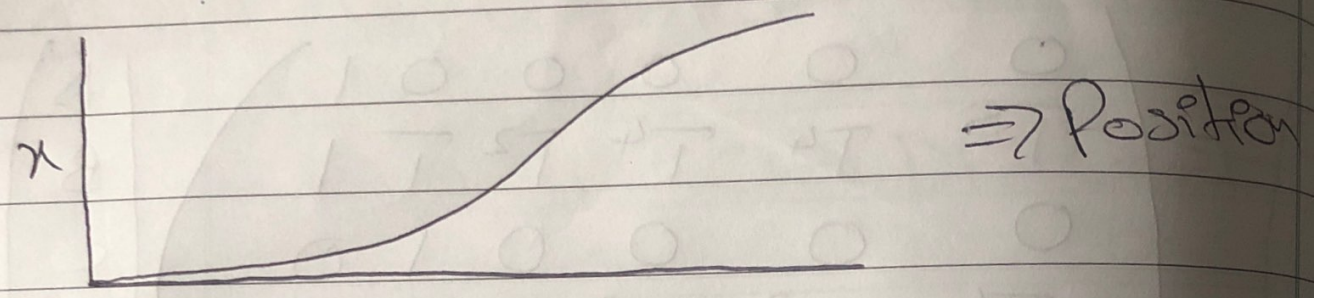
$$\begin{pmatrix} S_0 \\ S_T \\ S'_0 \\ S'_T \\ S''_0 \\ S''_T \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ T^5 & T^4 & T^4 & T^2 & T & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 5T^4 & 4T^3 & 3T^2 & 2T & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 20T^3 & 12T^2 & 6T & 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \\ E \\ F \end{pmatrix}$$

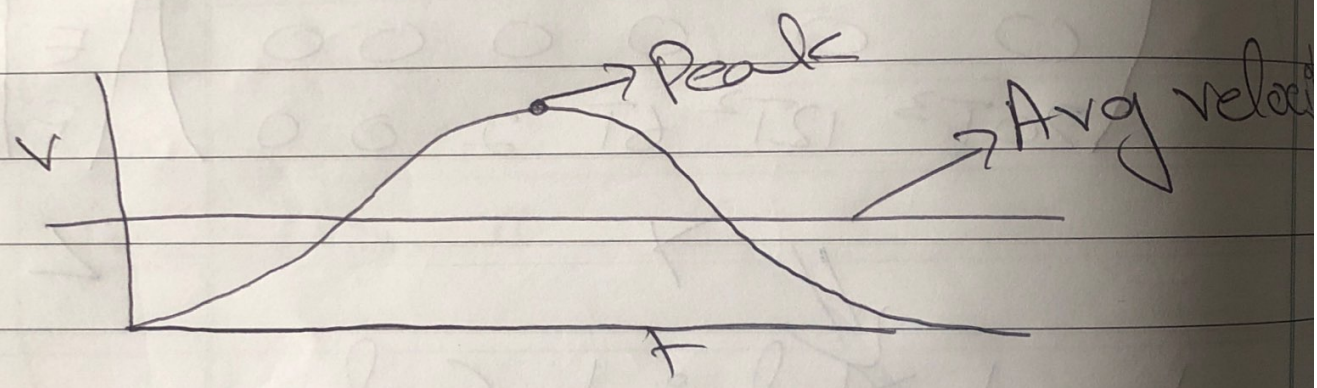
X = Y Z

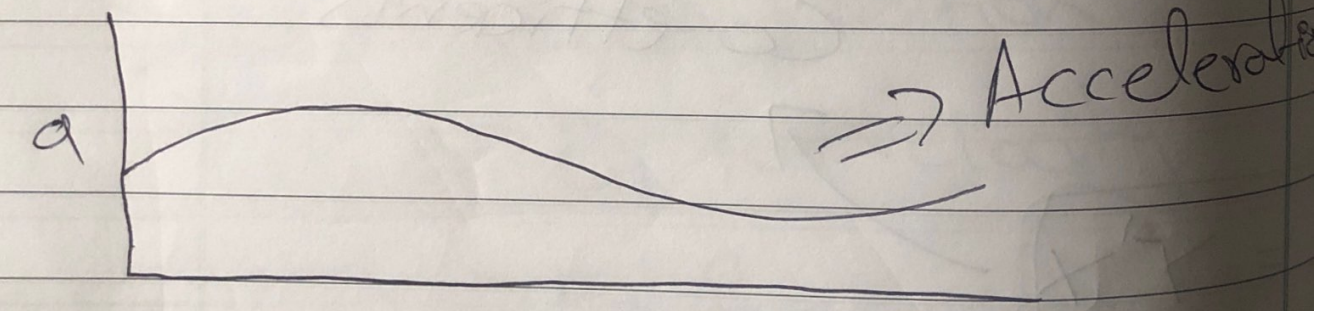
Invert Matrix to Get Co-efficients

$$Z = Y^{-1}X$$

Performance Problem



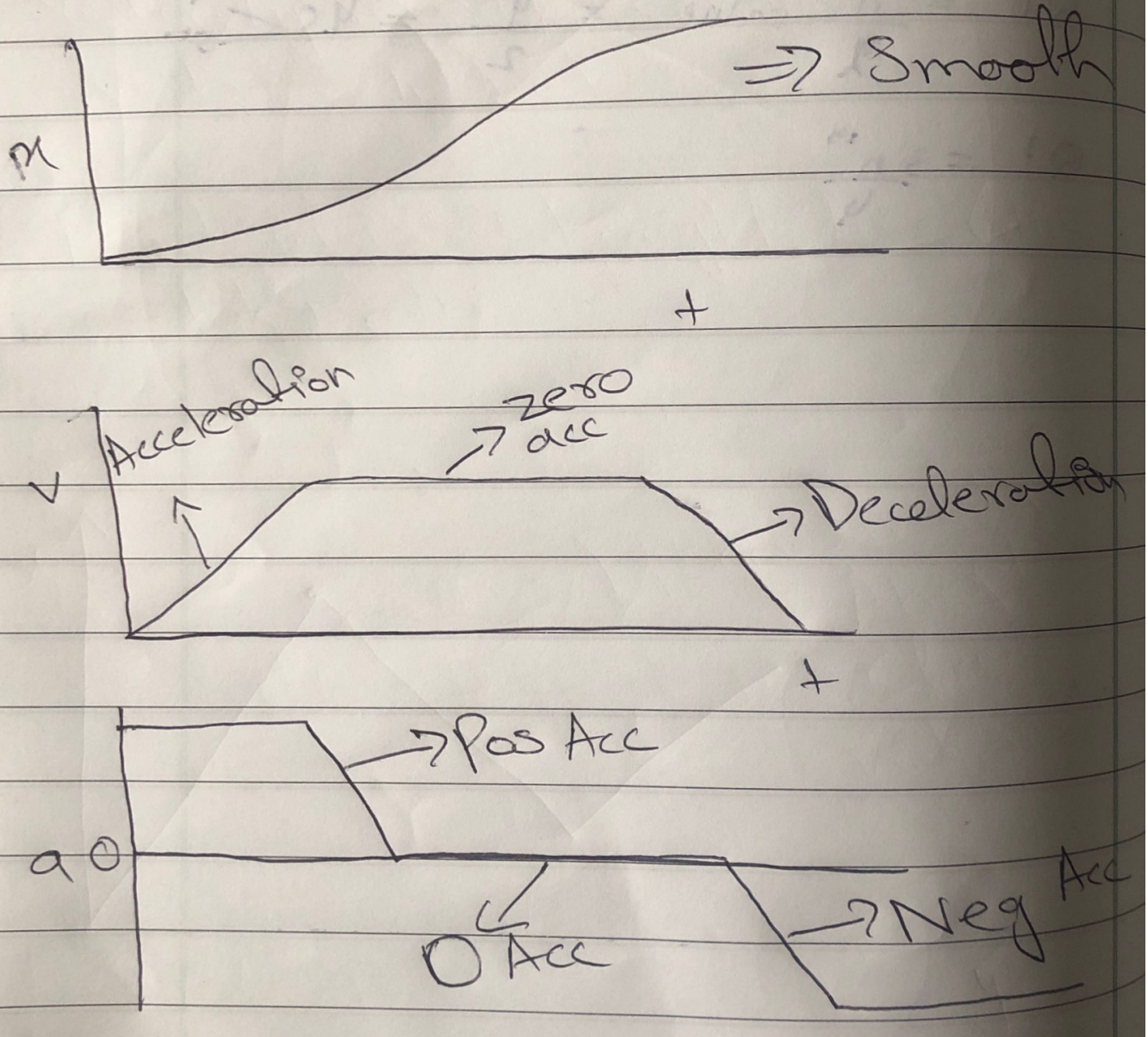




All are smooth

Problem: Avg velocity is only 52% of peak velocity

To fix this use Trapezoidal Profile



In the acceleration/deceleration phase the velocity increases linearly over time

The equation of velocity is

$v = at + b$

- a → const acceleration
- t → time
- b → initial velocity

The position follows a parabolic path

$s = at^2 + bt + c$

- at^2 → Rest same
- c → initial position

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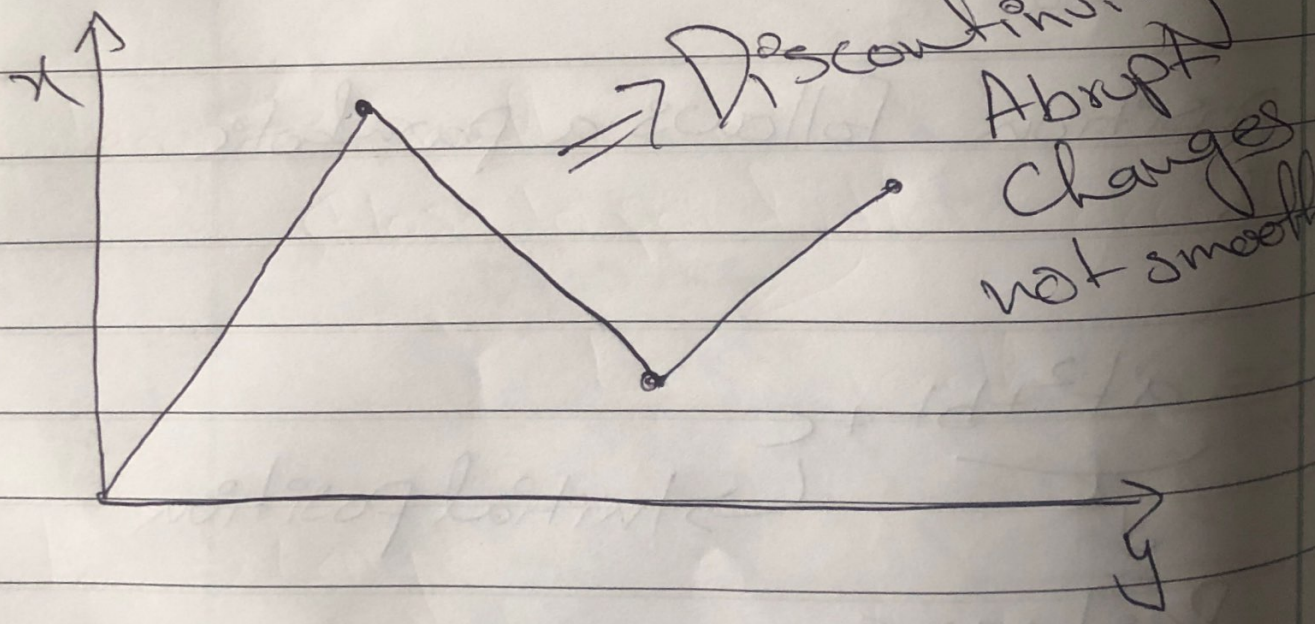
In the constant velocity phase

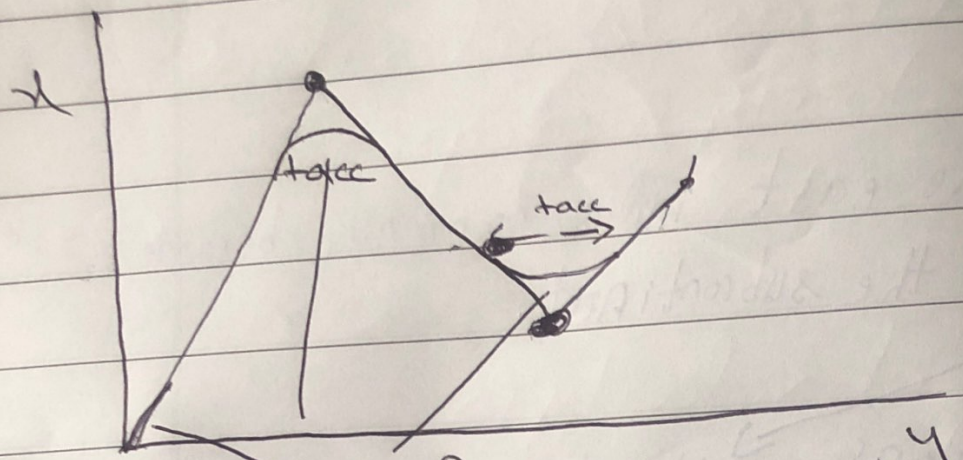
velocity remains constant

$v = b \longrightarrow$ Initial

Position is a straight line (linear)

$s = bt + c$





Blends
Smoothen out
the curve
Make smooth curve

t_{acc} ⇒ Small

Close to point, More accelera

↙

Less Smooth

t_{acc} ⇒ Large

Far from point, Less Acc

↙

More Smooth